

AYMANE AIT DADS

AI Data Science Summer Associate | Precision Medicine · Clinical ML Pipelines · Real-World Evidence Analytics

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PROFESSIONAL SUMMARY

End-to-end machine learning pipeline deployed at Orange Maroc processed 100K+ fiber-optic network records, cutting manual analysis time by ~60% through automated feature engineering, clustering, and stakeholder-ready Power BI outputs - directly demonstrating production-grade data science at scale. Ranked 1st in EURECOM class (0.791 AUC) on AI-generated image detection and placed top 10% globally in the NVIDIA Nemotron Reasoning Kaggle competition (\$106K+ prize pool, score 0.80/1.0), validating rigorous model evaluation and ablation methodology. Core competencies align directly with this role: Python, PyTorch, Scikit-learn, Hugging Face Transformers, SQL, ensemble methods, and real-world evidence analytics applied to high-stakes classification problems. Available for a Summer 2025 associate term; currently based in France and open to USA/Canada relocation, with eligibility to discuss visa sponsorship.

CORE COMPETENCIES

Real-World Evidence Analytics | Predictive Modeling | Feature Engineering | End-to-End ML Pipelines | Clinical Data Insights | Ensemble Methods | Stakeholder Communication | Model Evaluation & Ablation

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built end-to-end machine learning pipeline combining Random Forest classification and K-Means clustering on **100K+ fiber-optic network records** (upload speed, latency, jitter) to classify network quality across thousands of nodes.
- Developed a clustering pipeline mapping customer performance profiles to **5 commercial service tiers**, with output adopted directly by the network optimization team for infrastructure planning.
- Reduced manual analysis time by **~60%** through automated feature engineering, cross-validated model evaluation, and reproducible preprocessing workflows built in Python and Scikit-learn.
- Delivered stakeholder presentations translating predictive model outputs into actionable maintenance recommendations, visualized via **Power BI** dashboards for non-technical decision-makers.

PROJECTS

AI-Generated Image Detection - EURECOM

EURECOM Course + International Benchmark · 2025 · Ranked 1st in Class (0.791 AUC)

ImSecu + NTIRE 2026 @ CVPR

- Ranked **1st in EURECOM class leaderboard (0.791 AUC)** on a 250K-image binary classification task spanning 25+ generator types, then submitted independently to NTIRE 2026 @ CVPR CodaBench international benchmark.
- Fine-tuned CLIP ViT-L/14 (428M params) with a custom MLP head using dual learning rates, FP16 mixed precision, 10-view TTA, and a 3-model ensemble weighted by validation AUC, with ablation studies confirming 1-epoch training outperforms 4-epoch for OOD generalization.

PyTorch · OpenCLIP · Hugging Face Transformers · Kaggle T4 GPU · AdamW

NVIDIA Nemotron Reasoning Challenge -

Kaggle Global Competition · June 2026 · Top 10% Public Leaderboard · \$106K+ Prize Pool

Kaggle Competition (In Progress)

- Achieved **top 10% public leaderboard (0.80/1.0)** in a \$106K+ prize-pool competition by fine-tuning a 30B Mamba-Transformer MoE model with LoRA ($r=32$, ~888M trainable parameters) on 7,828 chain-of-thought reasoning examples using SFTTrainer with completion-only loss.
- Resolved **5 Blackwell GPU (sm_120) incompatibilities** in strict sequence - including OOM monkey-patching, Mamba CUDA kernel disabling, and Triton ptxas fixes - and ran 20+ single-variable ablations confirming packing and synthetic data as performance-degrading factors.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Statistics, Computer Vision, NLP, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | TRL / PEFT (LoRA, QLoRA) | Unsloth | TensorFlow | vLLM

Data Science & Analytics: Python | SQL | Pandas | NumPy | Feature Engineering | Ensemble Methods | Power BI | Clustering

MLOps & Infrastructure: Git | Linux | FP16/BF16 Mixed Precision | CUDA Debugging | Triton | Docker (basic) | Kaggle GPU | Ablation Studies